SEBASTIÁN ABELLO

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Education

Universidad Europea de Madrid 2022 – 2026
BSc in Aerospace Engineering (taught in English) *Madrid, Spain*

- **Relevant coursework**: Flight Mechanics, Structures, Aerodynamics, Fluid Mechanics, Satellite Design, Aircraft Design

University of Colorado Boulder
Specialization certificates in spacecraft dynamics and control on Coursera *Online*

- **Analytical Mechanics for Spacecraft Dynamics**: rigid-body kinematics and kinetics, Euler's equations, gravity-gradient torques and dual-spin spacecraft
- **Attitude Control with Momentum Exchange Devices**: Lyapunov-based nonlinear attitude control using reaction wheels and control moment gyroscopes (CMGs)

Experience

Airbus A400M Flight Simulator Mar. 2026 – Present
CT Ingenieros | Simulator Test & Integration *Getafe, Madrid*

- Supported **4 certification test campaigns** across different aircraft systems, with the **HIL** simulator ready for every session, by configuring, deploying and verifying software beforehand
- Minimized downtime during live test sessions by resolving issues in real time, leading simulator **troubleshooting** and flying the simulator during tests
- Validated software updates against real avionics per acceptance criteria by executing test procedures and acceptance tests
- Ensured full technical traceability of every session by documenting post-session reports, logs and discrepancy records
- Standardized session preparation with a 7-category checklist across 3 linked sheets by developing a **Python** (openpyxl) tool with formulas and conditional formatting

Projects

openLodestar | *Python, NumPy, pytest – Spacecraft attitude dynamics and control simulator* 2026 – Present

- Built an open-source attitude dynamics simulator, validated by **49 automated tests** across 3 GitHub releases, by implementing quaternion kinematics and Euler's equations
- Achieved closed-loop slew-and-hold maneuvers with **3 reaction wheels** by designing a **quaternion-feedback PD controller**
- Reproduced the **Dzhanibekov effect**, shown in a 3D animation, by modeling torque-free rigid-body dynamics

Roadster Demo Sim | *Python – 3D cold-gas thruster flight simulation* 2026

- Simulated the Tesla Roadster **cold-gas thruster** hop in 3D, cross-checked against hand-derived equations, by writing the control logic from scratch in **Python**

Thesis: SEPHAR | *OpenFOAM – Autonomous underwater glider for Europa (Jupiter)* 2026

- Designed the structure of an underwater glider for Europa's subsurface ocean, sized for extreme pressure and cryogenic conditions
- Validated the hull's hydrodynamic behavior under the ice through **CFD simulation in OpenFOAM**
- Enabled fully autonomous navigation by designing a center-of-gravity control system using a stepper-driven battery rod

CubeSat and Autonomous Explorer Rover | *SolidWorks, Python, C++*

- Achieved real-time yaw pointing correction on a CubeSat by integrating **IMU** sensors with a reaction wheel
- Achieved autonomous rover navigation with obstacle avoidance by programming the logic in Python and C++ and calibrating motor control

Technical Skills and Languages

Simulation & Analysis: ANSYS (Structural/Fluids), OpenFOAM, Simulink, technical data analysis
Programming: Python, C++, MATLAB (MathWorks Onramp), Arduino, VBA
CAD: SolidWorks (Advanced), Fusion 360, Onshape
Clearance & Nationality: Active NATO security clearance; dual Spanish (EU) and Colombian citizenship
Languages: Spanish (Native), English (C1/C2 – Bilingual), Portuguese (Intermediate)